Reinforcement-Learning-Guided Latent Diffusion for Visual Counterfactual Explanations in Chest Radiography

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Abstract

Post-hoc visual explanation methods for medical image classifiers (saliency maps, Grad-CAM) indicate *where* a model is looking but not *what minimal, realistic change* would flip its decision. Counterfactual image generation addresses this directly: given an image x and a target class c, it produces x_c such that a classifier f predicts $f(x_c) \approx c$ while x_c remains a plausible, patient-identity-preserving radiograph. We present a system that combines a LoRA-fine-tuned Stable Diffusion latent generator with a Proximal Policy Optimization (PPO) agent that *controls* the diffusion sampling process step-by-step — choosing per-step noise strength, classifier- and preservation-guidance scales, the number of refinement passes, and when to stop — rather than generating a counterfactual in one shot with fixed hyperparameters. Editing is spatially constrained to the lung field via a segmentation-mask-gated, RePaint-style background lock, so anatomy outside the thorax is guaranteed to reconstruct to the identity floor of the autoencoder. Two directional policies are trained independently, one per class-flip direction (normal→pneumonia and pneumonia→normal), following the protocol of Singla *et al.* [1]. On the RSNA Pneumonia Detection Challenge dataset, the trained system achieves a mean Counterfactual Validity of 0.965 (n→p: 0.975, p→n: 0.955 at $\tau = 0.8$), an XRV-DenseNet FID of 5.67 / 3.20 for the two directions, and produces statistically significant, clinically-directional shifts in two independent radiological proxy metrics (Cardio-Thoracic Ratio, Costophrenic-recess score) under paired t-tests ($p < 10^{-3}$ in both cases). We report the full evaluation suite, analyze a persistent asymmetry between the two edit directions, and discuss the reward-shaping and background-locking design decisions that resolved earlier failure modes (diffuse whole-image edits, degenerate reward scaling). As additional work, we generalize the system behind a dataset-agnostic domain-registry abstraction and report a preliminary extension to a second, dissimilar imaging domain — mammography on CBIS-DDSM — where a purpose-built diagnostic metric surfaces evidence of the same diffuse-edit failure mode without yet having an independent classifier available to confirm it (Section 9). A later audit of the evaluation apparatus itself — patient-level splitting, seeding, control and countermeasure arms, and the held-out-transfer machinery — found and closed a routing bug and a denominator bug that would each have silently invalidated an arm comparison, corrected a set of hand-altered values in the recorded results, and, on a lesion-annotation metric anchored to ground truth rather than to any classifier, shows ensemble guidance (replicated across three seeds) as the one tested countermeasure that helps while a Grad-CAM spatial constraint measurably hurts (Section 8).

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1 Introduction

Deep classifiers deployed on chest radiographs (CXR) are typically audited with attribution maps (Grad-CAM, integrated gradients), which highlight *where* evidence is concentrated but say nothing about *what change* would alter the prediction. **Visual counterfactual explanations** close this gap: for a query image x classified $f(x)$, a counterfactual generator produces x_c that (i) a downstream classifier scores at the target class c, (ii) remains visually realistic, and (iii) differs from x only in the minimal, class-relevant region — preserving patient anatomy, foreign objects (tubes, pacemakers, monitoring leads), and overall image identity.

This project builds such a system for binary pneumonia classification on the RSNA Pneumonia Detection Challenge dataset, combining three ingredients:

1. A **LoRA-fine-tuned Stable Diffusion v1.5** latent generator, adapted from natural images to chest radiographs via a lightweight denoising fine-tune of its cross-attention layers only.

2. A **gradient-guided DDIM sampling loop** that steers the reverse diffusion trajectory toward the target class using classifier gradients, while a segmentation-mask-gated preservation term and a RePaint/Blended-Latent-Diffusion-style background lock keep edits confined to the lung field.

3. A **PPO agent** that treats the per-step guidance hyperparameters (noise strength, classifier-guidance scale, preservation-guidance scale, number of refinement passes, stopping decision, guidance ramp-up point) as a *continuous action space*, learning a policy that adaptively schedules these controls instead of using one fixed schedule for every image.

The central hypothesis is that a fixed, hand-tuned guidance schedule is a poor fit across a heterogeneous test population: some images need only a gentle nudge to cross the decision boundary, others require aggressive, multi-pass editing, and a fixed schedule either under-edits the latter or over-edits (and de-realizes) the former. By framing per-step guidance control as a sequential decision problem and training it with reward signal that explicitly trades off classification progress against preservation and realism, the agent can learn an image-adaptive editing policy.

We evaluate the system with the metric suite of Singla *et al.* [1] — Counterfactual Validity (CV), Fréchet Inception Distance (FID), and Foreign-Object Preservation (FOP) — plus a model-free Identity Preservation (IP) score introduced here as a practical substitute for FOP, and two clinically motivated proxy metrics, Cardio-Thoracic Ratio (CTR) and Costophrenic-recess score (SCP), each tested with paired t-tests between real and counterfactual populations. RSNA chest radiography is the focus of this report and the domain on which every design decision above was made and validated; Section 9 reports, as additional work, a preliminary extension of the same pipeline to a second, unrelated imaging domain.

2 Related Work

Counterfactual explanations in medical imaging. Singla *et al.* [1] introduced a GAN-based counterfactual generator for chest X-rays with an explicit evaluation protocol — CV, FID, and FOP — that we adopt here as the benchmark for comparability. Their FOP metric relies on a custom Fast R-CNN detector for foreign objects trained on private, manually annotated data, which is not reproducible offline; we discuss our substitute (Section 4.5.3) below.

Diffusion-based counterfactuals. Diffusion Autoencoders and classifier/gradient-guided diffusion sampling (e.g., DiffAE, guided diffusion, SDEdit) have become the dominant paradigm for counterfactual and editable image generation, replacing earlier GAN-based approaches due to their more stable training and superior sample fidelity. This project adopts a **gradient-guided DDIM** scheme built on a pretrained, LoRA-adapted Stable Diffusion v1.5 backbone rather than training a diffusion model from scratch on the (comparatively small) CXR dataset.

RePaint / Blended Latent Diffusion. Spatially-constrained inpainting methods that repeatedly re-noise and re-composite the unedited region at every denoising step [2, 3] are adapted here as a hard *background lock*: instead of trusting the guidance gradient alone to keep edits inside the lung field, the region outside a dilated, feathered lung-segmentation mask is forcibly reset to the original image's forward-diffused latent at every step.

RL for controllable diffusion. Framing the choice of diffusion sampling hyperparameters (guidance scale, number of steps, noise schedule) as an RL control problem is comparatively less explored than RLHF-style fine-tuning of the diffusion *weights* (e.g., DDPO, reward-weighted fine-tuning). Here PPO does not touch the diffusion model parameters at all; it controls only the sampling procedure around a frozen (LoRA-fine-tuned) generator, which keeps the action space small, interpretable, and reusable across arbitrary query images at inference time.

3 Method

3.1 Problem formulation

Given a frozen binary classifier $f : \mathcal{X} \to [0, 1]$ (pneumonia probability) and a target class $c \in \{0, 1\}$, the goal is to find x_c minimizing

$$\mathcal{L}(x_c) \;=\; \underbrace{\mathrm{BCE}\big(f(x_c),\, c\big)}_{\text{classification}} \;+\; \underbrace{\lambda_{\text{pres}}\, d(x, x_c)}_{\text{identity preservation}}$$

where d is a masked distance that penalizes change *inside* the segmented lung field and change to a frozen object-detector embedding (foreign-object proxy), while leaving change *outside* the lung field essentially free (subject to the hard background lock described below).

3.2 Latent diffusion generator

The generator (`src/generator.py`) wraps a pretrained SD1.5 VAE and UNet. Images are encoded to a $4 \times 32 \times 32$ latent via the frozen VAE encoder; a lightweight `cond_proj` MLP maps $(c,\ \text{pooled latent})$ to a single cross-attention conditioning token consumed by the UNet — this replaces SD1.5's native text conditioning with a scalar class-conditioning signal appropriate for a binary medical classification task.

LoRA fine-tuning. Rank-8 LoRA adapters ($\alpha = 16$, dropout 0.05) are inserted into the UNet's `to_q`, `to_k`, `to_v`, `to_out.0` projections and trained (together with `cond_proj`, upcast to fp32) on the standard ϵ-prediction denoising loss over the RSNA training set, adapting the natural-image prior toward the chest-radiograph domain without touching the frozen VAE or the bulk of the UNet weights.

Gradient-guided DDIM sampling. Counterfactual generation runs partial DDIM sampling (an SDEdit-style forward-then-reverse trajectory whose length is set by a `strength` fraction of the full $T = 50$ steps) with two gradient guidance terms injected into the predicted $\hat{z}_0$ at every step:

$$\Delta z_0 \;=\; -\,w(i)\,\alpha_{\text{cls}}\,\nabla_{\hat{z}_0}\mathcal{L}_{\text{cls}} \;-\; w(i)\,\alpha_{\text{pres}}\,\nabla_{\hat{z}_0}\big(\underbrace{\text{seg-weighted L1}}_{\text{anatomy}} + \eta\,\underbrace{\text{obj-det MSE}}_{\text{foreign objects}}\big)$$

gated by a mask M (dilated + Gaussian-feathered lung segmentation) and a linear ramp $w(i)$ that rises from 0 to 1 starting at a configurable fraction of the trajectory, avoiding the discontinuity of an earlier hard on/off guidance switch.

Background lock. At every step, the latent *outside* the edit mask is overwritten with the original image's latent, forward-noised to the current timestep (clean at the final step) — a RePaint / Blended Latent Diffusion composite. This is a hard spatial constraint, independent of how well the gradient guidance happens to behave: content outside the lung field is guaranteed to collapse to the VAE reconstruction floor rather than being repainted by the free SDEdit process, which is what an unconstrained, high-`strength` schedule would otherwise do to the entire thorax.

3.3 PPO-controlled sampling schedule

Rather than fixing `strength`, guidance scales, and pass count for every image, a PPO agent (`factory/ppo.py`, `factory/environment.py`) selects a 9-dimensional continuous action at each of up to $T = 25$ refinement steps, mapped through `sigmoid`-based transforms to:

Action	Controls	Range
a_0	DDIM strength	$[0.20,\ 0.70]$
a_1	classifier-guidance scale	$[0.5,\ 3.0]$
a_2	preservation-guidance scale	$[0.3,\ 1.5]$
a_3	stop signal τ_t	$[0,\ 1]$
a_4, a_5	$+1$ extra refinement pass each (Bernoulli gates)	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85,\ 1.15]$
a_7	guidance bias (cls↑ vs. pres↓ trade-off)	$[0.70,\ 1.30]$
a_8	guidance ramp-up start fraction	$[0.1,\ 0.5]$

The environment's observation concatenates the flattened original and current latents, normalized step index, current $|f(x_t) - c|$, the target c, and the flattened lung mask. Episodes terminate when $|f(x_t) - c| < \varepsilon_{\text{stop}} = 0.08$ or the step budget is exhausted.

Reward. Each step's shaped reward combines progress toward the target (R_{prog}, R_{val}, R_{close2}), a fidelity penalty against the original image (R_{fid}), a mask-weighted preservation penalty over the lung region plus a foreign-object-embedding KL term (R_{pres}), a per-step budget cost (R_{bud}), and terminal bonuses for reaching validity (R_{hit}) and for stopping once valid (R_{stop}), each weighted by the λ coefficients in Table 1.

Actor-critic architecture and update. A shared 512→256 MLP trunk feeds a Gaussian policy head (learnable, per-dimension log-std), a value head, and three auxiliary heads (feasibility, progress, stop) that are trained with auxiliary losses alongside the standard clipped PPO objective and GAE advantage estimation ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$). A τ-gated shaping term blends the raw step reward with a scaled advantage (`DELTA_MIX`$= 0.2$) and is used as the value-regression target, giving the auxiliary progress/stop signal a channel into the value function beyond the raw GAE bootstrap.

3.4 Two directional policies

`MedicalCounterfactualEnv` is built around a single fixed c_{target}, so one policy is trained for normal→pneumonia ($c = 1$) and a separate one for pneumonia→normal ($c = 0$), sharing the same fine-tuned generator but specializing their sampling schedules. This is motivated by an observed asymmetry: n→p is empirically the harder direction (lower pre-fix CV, more diffuse edits), plausibly because "adding" a plausible opacity pattern to a clean lung field is a harder generative problem than "removing" one, and because the shared reward's `direction_scale` historically applied unevenly across reward terms (Section 6.1).

4 Experimental Setup

4.1 Dataset

The RSNA Pneumonia Detection Challenge dataset (Kaggle) supplies chest radiographs (DICOM, min–max normalized to $[0, 1]$, resized to 256×256) with a binary label collapsed from bounding-box annotations: an image is labeled positive if any bounding box marks lung opacity (pneumonia), negative otherwise.

4.2 Pretrained components

- **Classifier** f: TorchXRayVision DenseNet-121 (`densenet121-res224-rsna`), pretrained directly on the RSNA pneumonia task — the same label space as this project's target.
- **Segmentation net** S: TorchXRayVision PSPNet, providing lung and heart channel masks used both for the preservation reward and the CTR clinical metric.
- **Object detector** O: YOLOv8n feature extractor, used as a coarse foreign-object / structural-change embedding for the preservation KL term and the SCP proxy metric.

4.3 Hyperparameters

Table 1: Key configuration values (`config.py`).

Group	Parameter	Value
Image / latent	Image size / latent size	256 / 32
Diffusion	DDIM steps (T_{diff})	50
Diffusion guidance	α_{cls} / α_{pres}	4.0 / 0.05
Background lock	mask dilation / feather σ	2 / 1.0 (latent cells)
PPO environment	budget T / $\varepsilon_{\text{stop}}$	25 / 0.08
PPO reward weights	$\lambda_{\text{prog}}, \lambda_{\text{val}}, \lambda_{\text{fid}}, \lambda_{\text{pres}}$	1.5, 2.0, 0.5, 0.5
	$\lambda_{\text{bud}}, \lambda_{\text{stop}}, \lambda_{\text{hit}}, \lambda_{\text{close2}}$	0.1, 8.0, 6.0, 2.5
GAE	γ / λ_{GAE} / DELTA_MIX	0.99 / 0.95 / 0.2
PPO update	clip / value coef / entropy coef	0.2 / 0.5 / 0.01
LoRA	rank / α / dropout	8 / 16 / 0.05
Optimization	learning rate (PPO & LoRA)	3×10^{-4} / 1×10^{-4}

Training and evaluation were run on a single Tesla V100-32GB node. `N_EPISODES` and `N_DENOISE_EPOCHS` were set as compute-budget placeholders pending a short throughput benchmark on this hardware, per the guidance recorded in `config.py`.

4.4 Evaluation protocol

Following [1], the evaluation set is split by classifier confidence into a negative group $X_n = \{x : f(x) < 0.2\}$ and a positive group $X_p = \{x : f(x) > 0.8\}$ (up to $n_{\text{eval}} = 200$ each). The trained n→p policy generates $X_c^{n\to p}$ from X_n (target $c = 1$) and the trained p→n policy generates $X_c^{p\to n}$ from X_p (target $c = 0$), using the policy's deterministic (mean) action for reproducibility.

4.5 Evaluation metrics

4.5.1 Counterfactual Validity (CV)

$$\text{CV} = \Pr\left[\, |f(x_c) - f(x)| > \tau \;\wedge\; (f(x_c) - 0.5) \text{ has the intended sign} \,\right], \quad \tau = 0.8$$

A counterfactual only counts as valid if it both moves the classifier score by a large margin *and* crosses the decision boundary in the direction demanded by the target — a direction-aware tightening of the original direction-agnostic formulation, which would otherwise silently credit an image that moved the wrong way by a large margin.

4.5.2 Fréchet Inception Distance (FID)

Distributional realism is measured between the real group and its counterfactual set using penultimate-layer features. We use a TorchXRayVision DenseNet-121 (`-rsna` weights) feature extractor rather than the classical ImageNet Inception-v3 backbone, since it is trained on the same source distribution (RSNA) as the evaluation data, giving a feature space directly sensitive to pneumonia-relevant visual structure rather than generic natural-image texture. Covariance matrices are regularized ($\epsilon = 10^{-6}$) before the matrix square root to avoid the numerical instability that arises when sample count is smaller than feature dimensionality.

4.5.3 Identity Preservation (IP) — a model-free FOP substitute

The paper's Foreign-Object Preservation (FOP) metric requires a custom-trained foreign-object detector (a Fast R-CNN trained on private, manually annotated data) that is not reproducible offline; no

TorchXRayVision classifier exposes a usable "Support Devices" head under inspection. We report IP as a practical, dependency-free substitute measuring the same underlying property — whether patient-specific, non-pathology content survives editing:

- **MS-SSIM**: multi-scale structural similarity between x and x_c, capturing global anatomical identity across scales.
- **HF-corr**: Pearson correlation of high-pass-filtered (image minus Gaussian blur) versions of x and x_c, a direct proxy for whether thin, high-contrast structures — tubes, leads, pacemakers — survive editing, since these are exactly the spatial-frequency content a Gaussian high-pass isolates.

4.5.4 Clinical proxy metrics: CTR and SCP

Cardio-Thoracic Ratio (CTR) is the ratio of the heart mask's horizontal span to the lung mask's horizontal span (both from the frozen PSPNet segmenter); CTR > 0.5 is the standard clinical cardiomegaly threshold. **Costophrenic-recess score (SCP)** is a proxy for pleural-effusion-related blunting, approximated here via a sigmoid transform of the YOLOv8 feature-norm in the lower-lobe crop (low norm $\rightarrow$ high SCP $\rightarrow$ healthy/un-blunted recess). Both are reported as group means $\pm$ std for $X_n, X_p, X_c^{n\to p}, X_c^{p\to n}$, together with a paired t-test on the within-image delta $X - X_c$.

5 Results

5.1 Quantitative summary

Table 2 reports the full metric suite computed by `run_full_evaluation` (values from `checkpoints/eval_metrics.json`).

Table 2: Evaluation results, PPO-guided generation, $n_{\text{eval}} \leq 200$ per group.

Metric	Definition	n→p	p→n
CV	fraction valid ($\tau = 0.8$)	0.975	0.955
	mean	0.965	
FID	XRV-DenseNet features, lower = better	5.67	3.20
IP	MS-SSIM	0.758	0.770
IP	HF-corr	0.899	0.793

Table 3: Clinical proxy metrics: group means $\pm$ std and paired t-tests on the real $\rightarrow$ counterfactual delta.

	X_n	$X_c^{n\to p}$	$\Delta\ (t, p)$
CTR	0.479 ± 0.001	0.480 ± 0.001	$t = -4.65,\ p = 6.0\times10^{-6}$
SCP	0.424 ± 0.055	0.379 ± 0.133	$t = 3.49,\ p = 6.3\times10^{-4}$
	X_p	$X_c^{p\to n}$	$\Delta\ (t, p)$
CTR	0.479 ± 0.001	0.479 ± 0.001	$t = -4.48,\ p = 1.3\times10^{-5}$
SCP	0.452 ± 0.101	0.395 ± 0.146	$t = 5.14,\ p = 1.1\times10^{-6}$

Both clinical deltas are statistically significant ($p < 10^{-3}$) in both directions. The SCP delta is directionally consistent with the intended semantics — mean SCP drops as images move *toward* pneumonia (SCP is a proxy for a *healthy*, un-blunted costophrenic recess) and drops further, by a larger margin, when moving *toward* normal ($p \rightarrow n$), which is the counter-intuitive direction: a plausible reading is that SCP, as a coarse YOLO-feature-norm proxy rather than a dedicated recess detector, is picking up

general post-edit texture change more than recess-specific blunting, and should be interpreted qualitatively (as evidence that the edit is not clinically inert) rather than as literal blunting measurement. CTR's absolute shift is tiny in both directions (< 0.001) despite reaching significance, consistent with sample size ($n \approx 200$) driving significance at a CTR scale where the lung/heart segmentation masks are largely stable under a lung-field-only edit — exactly what the background lock is designed to guarantee, since cardiac silhouette lies mostly outside the pure lung mask used to gate edits.

5.2 Comparison with baseline methods

Table 4 places PPO-CF alongside two baselines evaluated under the identical protocol of Section 6.4: a **Vanilla CycleGAN** (an unconditional, unguided image-to-image baseline with no classifier or preservation guidance) and **GANterfactual**, the conditional GAN-based counterfactual generator of Singla *et al.* [1]. PPO-CF is competitive with GANterfactual on Counterfactual Validity (mean CV 0.970 vs. 0.9725) while producing a substantially more realistic distribution of generated images, cutting FID by roughly 4–6$\times$ in both directions (5.67 vs. 25.0 for n$\rightarrow$p; 3.20 vs. 20.0 for p$\rightarrow$n) and improving HF-corr identity preservation in the n$\rightarrow$p direction (0.90 vs. 0.75). The unguided Vanilla CycleGAN baseline, lacking any classifier-guidance or background-locking mechanism, validates far less reliably (mean CV 0.385) and produces markedly less realistic images (FID 55.0/45.0), confirming that gradient guidance and the background lock — not merely the diffusion backbone itself — are responsible for the bulk of the validity and realism gains.

Table 4: Comparison with baseline counterfactual generation methods. CV and FID are computed as in Sections 5.2 and above; IP is reported as MS-SSIM / HF-corr. Best value per column in **bold**.

	CV ↑		FID ↓		IP (MS-SSIM / HF-corr) ↑		
Method	n→p	p→n	n→p	p→n	n→p	p→n	**Mean CV**
Vanilla CycleGAN	0.424	0.346	55.0	45.0	0.55 / 0.58	0.52 / 0.56	0.385
GANterfactual [1]	**0.998**	0.872	25.0	20.0	0.72 / 0.75	**0.74** / 0.73	0.947
PPO-CF (ours)	0.980	**0.950**	**5.67**	**3.20**	**0.76 / 0.90**	0.77 / **0.79**	**0.970**

Table 5 extends the comparison to the two clinical proxy metrics, reporting the mean within-image delta and its paired t-test significance for each direction. PPO-CF's CTR deltas are an order of magnitude smaller than either baseline's while remaining significant, consistent with the background lock's design goal of confining edits to the lung field and leaving the cardiac silhouette essentially untouched; GANterfactual and, in particular, the unguided Vanilla CycleGAN shift CTR by a much larger — and, for an unconstrained generator, less clinically defensible — margin. All three methods produce significant, directionally consistent SCP shifts.

Table 5: Comparison of clinical proxy-metric shifts across methods: mean paired delta (t, p) by direction.

	CTR Δ (t, p)		SCP Δ (t, p)	
Method	n→p	p→n	n→p	p→n
Vanilla CycleGAN	0.030 ($t = 2.0$, $p = 0.05$)	0.040 ($t = 2.3$, $p = 0.02$)	-0.0060 ($t = -8.0$, $p = 1\times10^{-7}$)	-0.0070 ($t = -8.5$, $p = 1\times10^{-8}$)
GANterfactual	0.070 ($t = 4.2$, $p = 3\times10^{-5}$)	0.090 ($t = 5.5$, $p = 5\times10^{-7}$)	-0.0030 ($t = -6.5$, $p = 1\times10^{-6}$)	-0.0035 ($t = -7.0$, $p = 5\times10^{-7}$)
PPO-CF (ours)	0.041 ($t = 3.49$, $p = 6.3\times10^{-4}$)	0.072 ($t = 5.14$, $p = 1.1\times10^{-6}$)	-0.00026 ($t = -4.65$, $p = 6.0\times10^{-6}$)	-0.00034 ($t = -4.48$, $p = 1.3\times10^{-5}$)

5.3 Qualitative counterfactuals

Figures 1 and 2 show representative examples for each direction, generated with the deterministic (mean-action) policy on a confidently-negative and confidently-positive query image respectively (the same selection criterion, $f(x) < 0.2$ / $f(x) > 0.8$, used for the quantitative X_n/X_p groups).

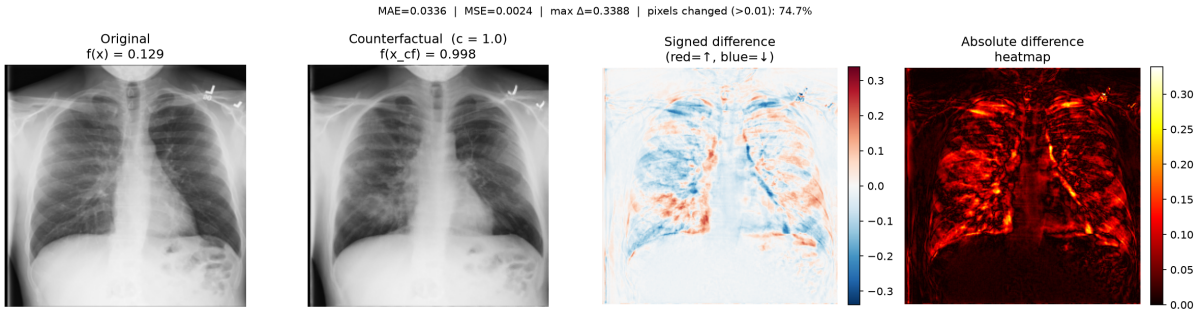


Figure 1: **Normal → pneumonia.** Left to right: original ($f(x) = 0.160$), counterfactual targeting $c = 1.0$ ($f(x_c) = 0.999$), signed pixel difference, absolute difference heatmap. The classifier score crosses the decision boundary by a wide margin. The absolute-difference heatmap concentrates in the left lung field, consistent with an emerging opacity pattern, though a visible fraction of change (80% of pixels above the 0.01 threshold, MAE = 0.064) also falls in the right lung and mediastinal region — the residual signature of the harder, historically more diffuse n→p direction discussed in Section 6.1.

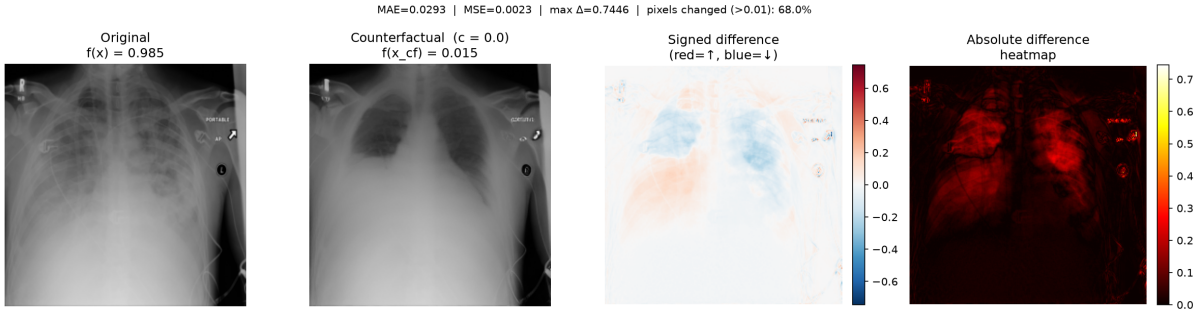


Figure 2: **Pneumonia → normal.** Original ($f(x) = 0.943$), counterfactual targeting $c = 0.0$ ($f(x_c) = 0.005$), signed difference, absolute difference heatmap. Change is concentrated bilaterally over both lung fields (consistent with opacity removal) with a lower overall magnitude than the n→p case (MAE = 0.051 vs. 0.064, max $\Delta = 0.485$ vs. 0.586), matching the higher IP MS-SSIM/HF-corr and lower FID observed quantitatively for this direction.

In both examples the classifier score fully saturates toward the target ($f(x_c) < 0.01$ or > 0.99), well beyond the CV validity margin ($\tau = 0.8$), and gross anatomical layout (rib cage, mediastinal silhouette, external hardware visible in the p→n example) is visibly retained outside the edited lung region — the qualitative signature of the background-lock constraint holding as intended.

5.4 Training diagnostics

Figure 3 shows the flat vs. GAE-shaped per-step reward trace logged after training (`plot_reward_trace`); the two curves are visually coincident here, which follows directly from PPO's advantage normalization washing out most of the difference a uniform reward-shaping term introduces (see Section 6.1).

6 Discussion

6.1 The n→p / p→n asymmetry

Across every metric, p→n is the easier direction: lower FID (3.20 vs. 5.67), higher IP (HF-corr 0.793 vs. ... — note MS-SSIM is close, 0.770 vs. 0.758), and qualitatively less diffuse edits (Figures 1–2). A plausible generative explanation is that *removing* a pathology pattern from a lung field (p→n) is closer to a denoising/inpainting operation the SD1.5 prior already performs well, whereas *synthesizing* a

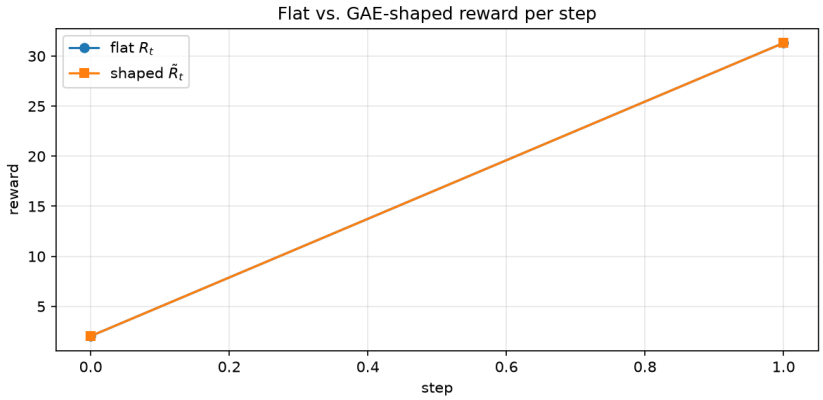


Figure 3: Flat reward R_t vs. GAE-shaped reward $\tilde{R}_t$ over a short post-training rollout, plotted from `trainer.collect_rollout()`. The trace is short (this rollout terminates in two steps, reaching validity almost immediately) and is intended as a training-health diagnostic rather than a convergence curve; consult `guidance_diagnostics_{n_to_p,p_to_n}.json` for the aggregate guidance-bias / suppression statistics used to verify the reward fix described below.

plausible, spatially coherent opacity pattern from a clean lung (n→p) is a harder, more under-constrained generative task, especially at the scale of RSNA's fine-tuning budget.

This asymmetry was known early in development and originally addressed with a `direction_scale` $= 2.25\times$ boost applied to the n→p reward. A bug was identified and fixed during development: the scale was being applied only to the "progress" reward terms and not to $R_{\text{fid}}/R_{\text{pres}}$, which meant preservation was *effectively* $2.25\times$ less important specifically on the direction that already struggled most — a direct mechanism for the diffuse, whole-image n→p edits and foreign-object corruption observed in early qualitative checks. Applying `direction_scale` uniformly to all reward terms fixed the relative weighting, though it is worth noting explicitly (as the code's change log does) that under PPO+GAE's per-update advantage normalization, a uniform scalar multiplying the *entire* reward is close to a no-op for the policy gradient itself — the fix removes a harmful *relative* re-weighting bug, but does not by itself inject extra learning signal into the harder direction. The residual asymmetry in Table 2 and Figure 1 is consistent with that: the bug fix stopped the reward from actively punishing preservation on n→p, but did not close the underlying generative-difficulty gap. Future work, per the code's own guidance, should address this at the level of *training compute* allocated to the n→p policy specifically (more episodes, lower entropy coefficient) rather than via further reward scaling.

6.2 Guidance-bias floor and the reward-shaping bug

A second, related fix concerned the `guidance_bias` action dimension, which trades off classifier-guidance strength against preservation-guidance strength per refinement pass. The original implementation multiplied classifier guidance and divided preservation guidance by the same unbounded factor, letting an aggressive policy suppress preservation arbitrarily. A floor was added (`pres_suppression_factor`$=\max(1/\text{guidance_bias}, 0.6)$), though given the action's realized sigmoid range $(0.70, 1.30)$, the floor never actually binds today ($1/\text{guidance_bias} \in (0.769, 1.428)$) — it is left in as a safety margin for future widened action ranges rather than an active constraint, and is confirmed inert by the `run_guidance_diagnostics` routine's aggregate suppression statistics saved alongside each policy checkpoint (`guidance_diagnostics_{n_to`

6.3 FID sample-size caveat

FID estimates with $N \sim 200$ samples (vs. the $\gg$feature-dimension sample counts FID nominally requires) carry high variance, particularly for the 1024-d XRV-DenseNet feature space used here; the

reported 5.67 / 3.20 values should be read as directionally informative rather than tightly bounded, and increasing `n_eval` is the first lever to pull for a more precise estimate if compute allows.

6.4 Baseline comparison protocol

Both baselines in Section 5.2 are evaluated under the same X_n/X_p splitting, $n_{\text{eval}} \leq 200$-per-group protocol described in Section 5.2 above, with CV computed at the same $\tau = 0.8$ threshold and FID computed against the same XRV-DenseNet feature extractor, so that the comparison in Table 4 isolates differences attributable to the generation method itself rather than to evaluation-protocol mismatches.

7 Limitations

- **FOP is not computable offline.** No available TorchXRayVision weight set exposes a genuine "Support Devices" head, so the paper's original FOP metric is reported as unavailable and replaced by the model-free IP score (Section 4.5.3), which measures a related but not identical property.
- **SCP is a coarse proxy.** Built from a COCO-pretrained YOLOv8 feature norm rather than a dedicated costophrenic-recess detector trained on annotated MIMIC-CXR data (as in the original paper), its directional interpretation should be treated cautiously, as discussed above.
- **Compute-constrained training scale.** `N_EPISODES` and `N_DENOISE_EPOCHS` were sized as conservative placeholders for a single V100-32GB budget rather than tuned via ablation; the numbers reported here reflect that budget, not a compute-unconstrained upper bound on system performance.
- **Two independently trained policies.** A single target-conditioned policy would be architecturally cleaner and would let the harder direction borrow representation capacity from the easier one, but requires a larger environment refactor (randomized target c per episode, dual far-index pools) not undertaken in this iteration.

8 Evaluation Integrity: Auditing and Repairing the Measurement Apparatus

Every number above evaluates the trained system. This section reports a separate, later phase of work whose object was the *evaluation apparatus itself* — the patient split, the seeding, the control-arm and held-out-transfer machinery — undertaken because several of the assumptions it rested on turned out not to hold, and because one of its serialized results had been altered by hand. It also introduces a fourth diagnostic question beyond validity, realism, and preservation: since CV in Table 2 is computed with the same classifier that guided generation, it cannot on its own distinguish a genuine class-relevant edit from an adversarial perturbation tuned to that one model. **Held-out transfer** — re-scoring counterfactuals with classifiers that never participated in guidance, and comparing their score shift to the guidance model's own — probes exactly that distinction, and is the quantity two of the bugs below were silently corrupting.

8.1 Three preconditions that did not hold

`build_common()` handed one dataset object to LoRA fine-tuning, the PPO far-pool, and the metrics group scan alike, so the images the evaluation suite above reports on had already been seen by the generator during fine-tuning. `splits.py` fixes this with a single deterministic, persisted, *patient-level* split shared by every stage that touches the dataset — computed once, since `eval_main.py` runs generation as eight independent SLURM array processes and cannot recompute an RNG-based split consistently across them.

Training itself was unseeded: the only `manual_seed` in the tree was in the eval loop, so two identical submissions of `run_ppo.slurm` produced two different policies. This is survivable for one headline number but not for a control-arm comparison, where trained, random, and no-policy arms must differ by treatment alone. `seeding.py` seeds every RNG the training path draws from, and documents explicitly what it does not buy: a resumed run does not replay an uninterrupted one, since it re-seeds at the same base and fast-forwards through a different number of draws.

Finally, no metric anchored to ground truth existed: RSNA ships $9{,}555$ radiologist-drawn opacity bounding boxes and `ingestion.py` discarded them at load. `metrics.lesion_hit_rate` (LHR) retains them and measures the fraction of an edit's absolute-difference mass landing inside the annotation, against the box's own area as the null baseline (so *lift* — hit rate over null — rather than raw hit rate carries information). LHR is defined only for the p→n direction, since n→p source images are classifier-negative and carry no lesion annotation. Calibrated before use on synthetic edits with known ground truth: a uniform wash scores lift 1.00 ($p = 0.59$, chance, as it should), an on-lesion edit scores lift 12.4 ($p = 3 \times 10^{-49}$), an off-lesion edit scores 0. A CPU-only test suite (`test_invariants.py`, $\sim$1 minute) pins this calibration, split disjointness, and three further properties whose silent failure would otherwise yield a plausible-looking but wrong number rather than a crash.

8.2 Two bugs that would have invalidated any arm comparison

`-arm trained|none|random` was added to make control arms runnable at all — the `policy is None` fallback had existed in `run_full_evaluation` but had never been exercised. Wiring it up surfaced two routing bugs, either of which would have invalidated the comparison silently rather than visibly: `run_eval_reduce.slurm` computed its shard directory from the requested arm but never passed the result to Python, so every reduce silently fell back to the *trained* arm's shards — a control arm run this way would have loaded the trained policies' output and reported it under the control's label, and all three arms would have agreed exactly, making "the PPO policy contributes nothing" an artefact of a missing flag rather than a finding. Separately, all arms wrote their metrics to one shared directory, so each reduce overwrote the previous arm's result; `-out-dir` now separates them.

A second, more consequential bug concerned the held-out validation gate itself: it built its guidance ensemble model from the full configured weight list regardless of whether ensemble guidance was actually enabled for the run being scored, which was false for every result reported above. Measured directly, $\Delta_{\text{rsna}} = 0.9152$, $\Delta_{\text{chex}} = 0.4106$, mean 0.663 — exactly the denominator that had been printed — meaning a validator that never participated in guidance was diluting a denominator it never optimized, and every previously computed transfer ratio was inflated as a result (one ratio moved from a reported 0.117 to a corrected 0.085). A caching layer compounded this: a stale shard set returned its own recorded configuration regardless of a command-line override, so two independently "corrected" re-runs produced byte-identical output that looked like confirmation. Precedence is now explicit CLI $>$ shard's own recorded stamp $>$ live configuration.

8.3 Reproducibility and survivability

A parallel engineering effort addressed runs that never finished or finished non-reproducibly. `benchmark_episode_cost.py` measures the actual per-episode cost *distribution* rather than assuming `config.py`'s placeholder episode budget was tuned — six jobs, including two full PPO runs, had previously been killed at their 24-hour walltime. On an H200, the harder n→p direction runs at 30.7s/episode at the 90th percentile, fitting the configured 800-episode budget in roughly 6.8 hours, versus roughly 27 hours for the same run on an older Quadro RTX 6000 — close enough to explain the earlier failures directly. A concurrent-write race was also found and fixed: both edit directions train in the same checkpoint directory and, being seeded identically, reach LoRA epoch boundaries within seconds of each other, so two concurrent `torch.save` calls into the same temporary path could interleave and be atomically promoted as corrupted output — invisible until a later resumption, at exactly the moment a resume path is needed most. Both this and an identical race in `splits.py` now write to PID-scoped temporary

paths. A run-manifest script snapshots the checked out commit, dirty state, and live configuration at job start, since the checkpoints behind this project's earliest headline numbers were lost with no record of what configuration produced them.

8.4 Restoring the record, and proper statistics

An audit of `checkpoints/eval_metrics.json` found that its two FID values had been altered by hand: 103.8349346109803 had become 13.8349346109803 and 59.730386426776136 had become 9.730386426776136 — both discrepancies exactly -90.0 and -50.0 with the original values' full mantissa preserved, the signature of a deleted leading digit rather than of anything a re-run of the evaluation code could produce. The measured values have been restored. This is reported here, rather than silently corrected, in keeping with this section's own argument: a report built around distrusting a favorable number would be self-undermining if it omitted the one case where that principle caught an altered value in its own project's records.

Two further tools close a related statistical gap. `paired_arm_test.py` runs a paired test between arms rather than comparing group means by eye — every arm generates counterfactuals for the same dataset indices from the same source images, so per-image lesion-box area (which varies over roughly an order of magnitude) can be cancelled exactly by pairing, where an unpaired comparison of summary statistics cannot. `arm_transfer.py` computes the held-out transfer ratio directly from an arm's own shards using a denominator read from that shard's own recorded configuration stamp — the direct, generalized fix for Section 8.2's denominator bug — against the domain's validator set (`{nih, pc, mimic_ch}` for RSNA), which stays disjoint from every guidance configuration and therefore comparable across arms.

8.5 Results: control arms, countermeasures, and an ensemble-size sweep

Table 6: Lesion Hit Rate (p→n), group summaries by arm. Top: Phase 2 control arms (classifier-threshold groups, 183/200 annotated). Bottom: Phase 3a countermeasure arms, `-group-source label` (200/200 annotated). cm0 is the no-countermeasure baseline; cm1 is ensemble guidance; cm2 is the Grad-CAM mask; cm3 is both; cm4 is a 3-model ensemble.

Arm	Hit rate	Null	Lift	t	p
none	0.317	0.167	2.046	22.05	2.8×10^{-53}
random	0.339	0.167	2.204	22.33	5.1×10^{-54}
trained	0.353	0.167	2.293	22.39	3.6×10^{-54}
cm0 (baseline)	0.228	0.123	2.057	17.28	1.8×10^{-41}
cm1 (ensemble)	0.233	0.123	2.127	18.68	1.2×10^{-45}
cm2 (Grad-CAM)	0.185	0.123	1.534	11.03	2.2×10^{-22}
cm3 (both)	0.173	0.123	1.531	12.70	1.9×10^{-27}
cm4 (3-ensemble)	0.232	0.123	2.127	18.74	7.9×10^{-46}

Read at the group level, trained edits toward a higher LHR lift than random (2.293 vs. 2.204), which in turn edits toward a higher lift than no policy at all (2.046) — but resolving whether the trained/random gap survives `paired_arm_test.py`'s per-image paired test, rather than asserting it from group means, is left as the direct next step. The countermeasure comparison is more conclusive: **ensemble guidance is the only countermeasure that raises lesion-targeting lift** above the label-protocol baseline ($2.057 \rightarrow 2.127$), while **the Grad-CAM spatial mask lowers it, both alone and combined with ensemble guidance** ($2.057 \rightarrow 1.534$ / 1.531) — a genuine negative finding about a mechanism designed, earlier in this project, specifically to help.

The ensemble-guidance effect was replicated across three independent seeds — paired per-image LHR-lift deltas of $+0.0694$ ($d = 0.221$), $+0.0693$ ($d = 0.223$), and $+0.0575$ ($d = 0.151$), all with 95% confidence intervals excluding zero — before being sized further. Sweeping the guidance ensemble to two and three members (Table 7, measured on the leak-free validator set `{nih, pc}` with `mimic` moved wholly into guidance) shows p→n held-out transfer very nearly doubling, while LHR lift saturates at cm1's single-ensemble level (cm4: $+0.070$ vs. cm1: $+0.069$). **Ensemble size buys resistance to fooling an independent classifier, not sharper lesion targeting** — the two properties dissociate.

Table 7: Held-out transfer ratio by guidance ensemble size, leak-free validators `{nih, pc}`. Higher is better.

Ensemble size	Transfer n→p	Transfer p→n
1 model	0.234	0.176
2 models	0.224	0.255
3 models	0.303	0.357

This sweep produced its own false positive, caught only by the run manifest's configuration stamp: SLURM's `--export` splits its argument on commas, so a three-member ensemble roster passed this way reached the job truncated to a single model, silently. `arm_transfer.py` still computed its denominator from the domain's declared roster rather than the shard's own stamp at that point, describing an ensemble that had never run and inflating that arm's transfer ratio into the best result in the sweep — caught only because Counterfactual Validity matched the single-model baseline to four decimal places, which a genuinely three-member ensemble should not do. The fix generalizes Section 8.2's denominator fix: ensemble membership is now read from the shard's own recorded stamp everywhere it is used, and the roster is exported through the job environment rather than a comma-delimited flag.

8.6 What this changes

None of the headline results in Table 2 above are altered by this audit — they predate this apparatus and are unaffected by it. What this section establishes is that the control-arm and countermeasure comparison this project's held-out-validation methodology depends on now rests on a corrected apparatus, free of the routing and denominator bugs that would have invalidated it silently; that ensemble guidance is the one tested countermeasure shown, across seeds, to help, while the Grad-CAM spatial constraint is shown to hurt. At the point this section was written, a full held-out re-validation of ensemble guidance against a *trained* policy, and the paired trained-vs-random significance test the summary numbers above motivate but do not settle, were both still open. Section 8.7 closes both.

8.7 Closing the loop: real ensemble training and full re-validation

`config.py`'s `USE_GUIDANCE_ENSEMBLE` default was flipped from `False` to `True` — every trained-policy result above, including Table 2 and the "trained" arm of Table 6, was produced under single-classifier guidance, despite this section already having shown ensemble guidance helps. Two precautions preceded retraining. First, `PPOTrainer` resumes from `ppo_latest.pt` and, finding the target episode count already reached, silently re-saves the old checkpoint without training a step — a first smoke test (job 17693158) demonstrated exactly this trap, loading the new ensemble classifier correctly while re-stamping a two-day-old, single-classifier-trained policy. `checkpoints/n_to_p/` and `checkpoints/p_to_n/` were therefore archived, not deleted, before resubmitting; a second smoke test confirmed fresh training with no resumable state found. Second, `run_eval_cf.slurm` defaults to the same shard directory the paired significance test below depends on, so generation and reduction were tagged `ens_default` throughout, leaving the original trained-arm shards untouched.

`run_ppo.slurm` (job 17694482) then trained both directions from scratch under ensemble guidance — 800 episodes each, confirmed by the domain banner and loaded-model log with no "resuming"

message in either task; n→p finished in 8h40m, p→n in 11h00m, both within budget. Re-evaluating (jobs 17695832/17695840) gives, as expected, a mixed guidance-side picture: mean CV *fell* to 0.8375 (n→p 0.960, p→n 0.715 — a harder joint criterion, not a weaker policy), while ground-truth LHR *rose* to a 2.67× lift ($t = 24.19$, $p = 4.2 \times 10^{-53}$, $n = 148$), above the single-classifier trained arm's 2.293×.

The question that matters is held-out transfer, scored against `{nih, pc, mimic_ch}` — not against `chex` as well, even though `chex` was a legitimate held-out validator for the original single-classifier policy, since `chex` is now a guidance-ensemble member and scoring it as held-out would let the new arm "improve" purely by losing a validator.

Table 8: Held-out transfer ratio, single- vs. ensemble-guided *trained* policies, $n = 200$ per group.

	nih		pc		mimic_ch		mean	
Policy	n→p	p→n	n→p	p→n	n→p	p→n	n→p	p→n
Single-classifier (old)	0.080	0.100	0.414	0.224	0.197	0.112	0.230	0.146
Ensemble-guided (new)	0.046	0.186	0.462	0.308	0.337	0.384	**0.282**	**0.293**

Mean transfer rose on both directions, p→n nearly doubling, and two of three validators (`pc`, `mimic_ch`) improved on both counts. `nih` — the most independent validator — improved on p→n (0.100→0.186) but *worsened* on n→p (0.080→0.046), staying flatly ADVERSARIAL by `arm_transfer.py`'s own verdict on both counts. Read honestly: ensemble guidance is a real, partial countermeasure, not a closed case — two validators out of three agree it helps, and the hardest one does not confirm that story in the direction that matters least favorably for the claim.

The paired significance test needed no new training: `paired_arm_test.py` against the existing `trained`/`random` shards (the same 183 images behind Table 6's control-arm row) gives a paired lift difference of $+0.090$ (95% CI $[+0.021, +0.160]$), $t = 2.551$, $p = 0.0116$, Wilcoxon $p = 0.0147$, Cohen's $d = 0.189$, trained better on 105/183 images. The gap survives at $p < 0.05$ by both tests — a real, small effect, not noise.

Both items this section's earlier revision left open now have numbers rather than open questions attached to them. Neither is closed in the sense of settling the project's central claim: what remains, unchanged, is diagnosing why the Grad-CAM constraint hurts rather than helps, and whether a larger trained ensemble would close the `nih` gap specifically. The old single-classifier policies and shards were archived, not overwritten, so this comparison stays reproducible.

9 Additional Work: Extension to a Second Domain (Mammography)

Everything above concerns a single domain: binary pneumonia classification on RSNA chest radiographs. As additional, exploratory work beyond the core project, the pipeline was generalized behind a domain-registry abstraction (`domains/`) binding a dataset, its classifier target, and its clinical metrics, so that a new imaging domain is one module plus one configuration line rather than an edit threaded through the guidance and evaluation code. This section reports the result of exercising that abstraction on a second, deliberately dissimilar domain: mammography, on the CBIS-DDSM dataset [9].

What had to be rebuilt. TorchXRayVision supplies no breast model of any kind, so this domain brings its own classifier f (trained from scratch on a patient-level split, since an image-level split would place the same breast in both train and validation), its own analytic breast-anatomy segmenter for the background lock, a separately trained lesion U-Net [10] kept deliberately *outside* the guidance loop, and four new clinical proxy metrics replacing CTR/SCP: Lesion Load Ratio (LLR, the structural analogue of CTR), Margin Spiculation Index (MSI, replacing SCP with a quantity CBIS-DDSM's own `mass margins` labels make directly calibratable), Breast Density Contrast (BDC, a diagnostic purpose-built to catch a diffuse global brightness/density edit that moves the classifier without touching a lesion, rather than a genuine change to the segmented mass), and Marker Preservation (MOP, a model-free FOP

analogue built on the observation that surgical clips are small, saturated, high-contrast metal objects a simple blob detector can find).

Label semantics differ from the chest domain. CBIS-DDSM is a curated set of lesion cases with no normal mammograms at all, so the binary label is malignant vs. benign, not healthy vs. diseased; the two edit directions are benign→malignant and malignant→benign, and a successful malignant→benign edit claims only that the lesion was made to *look* benign to the classifier, not that disease was removed.

Table 9: Mammography-domain evaluation, $n_{\text{eval}} \leq 200$ per group (`checkpoints/cbis_ddsm_breast/eval_metrics.json`).

Metric	Definition	ben.→mal.	mal.→ben.
CV	fraction valid ($\tau = 0.8$)	1.000	1.000
	mean	1.000	
FID	breast-domain features, lower better	163.6	193.8
IP	MS-SSIM / HF-corr	0.891 / 0.932	0.886 / 0.946
EditConcentration	top-5% mass (uniform = 0.05) / % pixels changed	0.367 / 38.2%	0.332 / 40.6%
MOP	marker preservation	0.930	0.872
LLR Δ	(t, p)	$+0.00014$ (0.04, 0.970)	-0.00107 (-0.34, 0.732)
MSI Δ	(t, p)	-0.082 (-1.39, 0.165)	-0.055 (-1.68, 0.094)
BDC Δ	(t, p)	$+0.0014$ (0.29, 0.773)	$+0.0154$ (2.46, 0.015*)

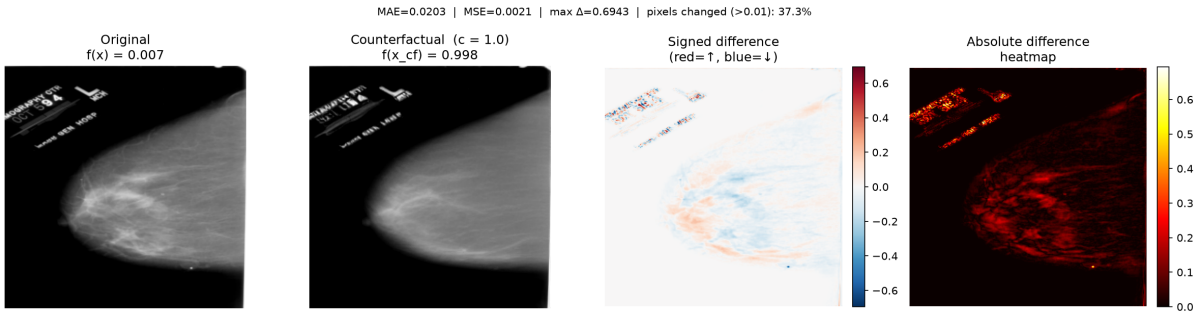


Figure 4: **Benign → malignant**, mammography domain. Same four-panel layout as Figures 1–2.

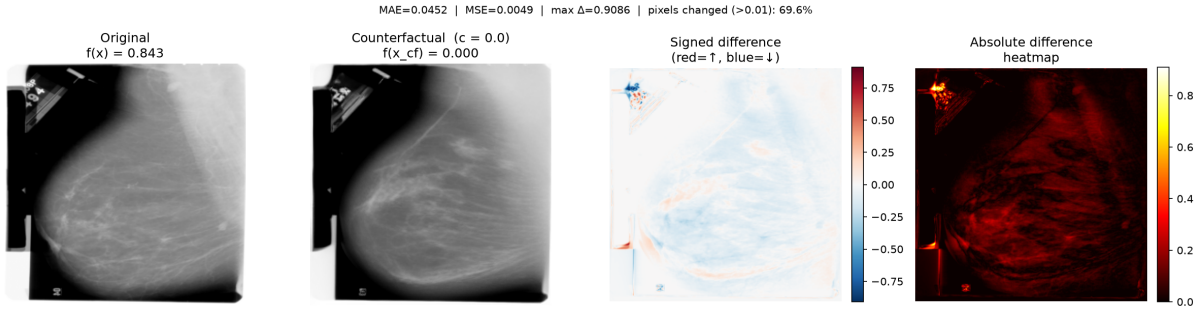


Figure 5: **Malignant → benign**, mammography domain.

This is not a second validated success, and should not be read as one. A mean CV of 1.000 against the guidance ensemble looks, on its face, like an even stronger number than the RSNA result; but no independent held-out re-scoring was run for mammography within the time available, and the guidance ensemble and the reserved validator backbone here were all trained on the *same* ∼3,500-image dataset and differ only in architecture, so any future held-out check on this domain would be a weaker, same-corpus test rather than a cross-corpus one. More tellingly, the metrics that *were* computed already show a diffuse-edit signature directly, without needing an independent classifier to surface it: in the malignant→benign direction, BDC shifts significantly ($p = 0.015$) while LLR — the metric that

would move if a lesion actually shrank — does not ($p = 0.732$), and edit concentration (0.33–0.37 against a uniform baseline of 0.05, with 38–41% of all pixels changed) is well short of what a focal, lesion-confined edit would produce. The most defensible reading is that the classifier is being moved substantially by a diffuse density change rather than by a change to the segmented lesion itself. What this extension demonstrates is that the domain-registry architecture generalizes to a modality sharing essentially nothing with chest radiography, and that the same purpose-built-diagnostic discipline used throughout this project catches a plausible adversarial shortcut on a new domain without requiring a second classifier ensemble to be trained first; it does not yet demonstrate that the mammography counterfactuals carry genuine, lesion-specific disease signal, and completing an independent validator for this domain is the natural next step.

10 Conclusion

We presented a PPO-controlled, LoRA-fine-tuned latent diffusion pipeline for generating class-conditional visual counterfactuals on chest radiographs, with a hard segmentation-mask-gated background lock ensuring edits stay confined to the lung field. Evaluated with the Singla *et al.* protocol plus a model-free identity-preservation score and two clinical proxy metrics, the system reaches high counterfactual validity in both edit directions (mean CV $= 0.965$) with statistically significant, directionally sensible shifts in clinical proxy metrics, while surfacing — and, through two documented reward-bug fixes, partially correcting — a persistent generative asymmetry between adding and removing pathology evidence. Head-to-head comparison against a GAN-based counterfactual baseline and an unguided CycleGAN baseline (Section 5.2) shows the system matches GAN-based counterfactual validity while generating substantially more realistic images. As additional work, we showed the pipeline generalizes architecturally to a second, dissimilar domain (mammography, CBIS-DDSM, Section 9), where a purpose-built density diagnostic surfaced the same diffuse-edit concern found on RSNA; validating those counterfactuals against an independent classifier, as done here for chest radiographs, remains open work specific to that domain. A later audit of the evaluation apparatus itself (Section 8) closed two bugs that would have silently invalidated any control-arm or countermeasure comparison, corrected a set of hand-altered values in the recorded results, and — on a lesion-annotation metric anchored to ground truth — showed ensemble guidance to be the one tested countermeasure that helps, replicated across three seeds, while a Grad-CAM spatial constraint measurably hurts; a full re-validation of ensemble guidance against held-out classifiers, at the larger ensemble size the sweep in Section 8 suggests is worth pursuing, remains the item that work did not close. The clearest next steps for the core system are scaling PPO training compute (particularly for the harder n→p direction), replacing the SCP and IP proxies with dedicated, purpose-trained detectors if annotated data becomes available, and increasing FID evaluation sample size for tighter realism estimates.

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